

MAINTENANCE

Engine

Drive belt tension						
Alternator	4A-FE		New belt.	160 ± 20 lbf		
			Used belt	130 ± 20 lbf		
	3S-GTE	w/ A/C	New belt	165 ± 10 lbf		
			Used belt	84 ± 15 lbf		
	5S-FE	w/o A/C	New belt	150 ± 25 lbf		
			Used belt	130 ± 25 lbf		
		w/ A/C	New belt	165 ± 10 lbf		
			Used belt	110 ± 10 lbf		
	w/o A/C	New belt	125 ± 25 lbf			
		Used belt	95 ± 20 lbf			
PS pump			New belt	125 ± 25 lbf		
A/C compressor (4A-FE)			Used belt	80 ± 20 lbf		
			New belt	160 ± 25 lbf		
			Used belt	100 ± 20 lbf		
Engine coolant capacity (w/ Heater)						
	4A-FE		M /T	5.2 liters	5.5 U S qts	4.6 Imp. qts
			A/T	5.6 liters	5.9 U S qts	4.9 Imp. qts
	3S-GTE			6.5 liters	6.9 US qts	5.7 I m p. qts
				6.2 liters	6.6 US qts	5.5 Imp. qts
	5S-FE		M /T	6.1 liters	6.4 U S qts	5.4 Imp. qts
			A/T			
Engine oil capacity (Drain and refill)						
4A-FE		w/ Oil filter change	3.2 liters	3.3 US qts	2.8 Imp. qts	
		w/o Oil filter change	3.0 liters	3.1 U S qts	3.4 Imp. qts	
3S-GTE		w/ Oil filter change	3.9 liters	4.1 U S qts	3.4 Imp. qts	
		w/o Oil filter change	3.6 liters	3.8 US qts	3.2 Imp. qts	
5S-FE	w/ Oil cooler	w/ Oil filter change	4.2 liters	4.4 U S qts	3.7 Imp. qts	
		w/o Oil filter change	3.8 liters	4.0 U S qts	3.3 Imp. qts	
	w/o Oil cooler	w/ Oil filter change	4.1 liters	4.3 US qts	3.6 Imp. qts	
			w/o Oil filter change	3.7 liters	3.9 U S qts	3.3 Imp. qts
Spark plug						
Type	4A-FE		ND	Q16R-U		
			NGK	BCPR5EY		
	3S-GTE		ND	PK20R8		
			NGK	BKR6EP8		
	5S-FE		ND	PK20R11		
			NGK	BKP6EP-11		
Gap	4A-FE and 3S-GTE		0.8 mm	0.031 in.		
			1.1 mm	0.043 in.		
Firing order			1 - 3 - 4 - 2			
Valve clearance	4A-FE	Intake	0.15 - 0.25 mm	0.006-0.010 in.		
		Exhaust	0.20 - 0.30 m m	0.008-0.012 in.		
	3S-GTE	Intake	0.15 - 0.25 m m	0.006 - 0.010 in.		
		Exhaust	0.28 - 0.38 m m	0.011 - 0.015 in.		
	5S-FE	Intake	0.19 - 0.29 m m	0.007 - 0.011 in.		
		Exhaust	0.28 - 0.38 mm	0.011 - 0.015 in.		
Idle speed (4A-FE)			800 rpm			

Chassis

Brake pads and disc				
Pad thickness		Limit	1.0 mm	0.039 in.
Disc thickness	Limit	Front	23.0 mm	0.906 in.
		Rear	9.0 mm	0.354
Disc runout	Limit	Front	0.07 mm	in.
		Rear	0.15 mm	0.0028 in.
Brake linings and drums				0.0059 in.
Lining thickness		Limit	1.0 mm	0.039 in.
Drum inside diameter	Limit	Drum brake	201.0 mm	7.913 in.
		Disc brake	171.0 mm	6.732 in.
Front axle and suspension				
Ball joint vertical play		Limit	0 mm	0 in.
Steering wheel play			30 mm	1.18 in.
Torque specifications				
Front seat mounting bolts			37 N-m	375 kgf-cm 27 ft-lbf
Engine mounting center member-to-body mounting bolts			73 N-m	740 kgf-cm 54 ft-lbf
Front suspension lower crossmember-to-body mounting bolts			152 N-m	1,550 kgf-cm 112 ft-lbf
Rear suspension lower crossmember-to-body mounting bolts			159 N-m	1,620 kgf-cm 117 ft-lbf

ENGINE MECHANICAL (4A-FE)

Specifications

Intake manifold vacuum	at idle speed		60 kPa	450 mmHg	17.7 in.Hg
Compression pressure	at 250 rpm	STD	1,320 kPa (13.5 kgf/cm ² , 191 psi) or more		
	Limit		981 kPa 10.0 kgf/cm ² , 142 psi		
	Difference of pressure between each cylinder		98 kPa (1.0 kgf/cm ² , 14 psi) or less		
Idler pulley tension spring	Free length		38.4 mm	1.512 in.	
	Installed load at 51.9 mm (2.043 in.)		35 – 39 N	3.6 – 4.0 kgf 7.9 – 8.8 lbf	
Cylinder head	Warpage Cylinder block side	Limit	0.05 mm	0.020 in.	
	Manifold side	Limit	0.10 mm	0.039 in.	
	Valve seat Refacing angle		30°, 45°, 60°		
	Contacting angle		45°		
	Contacting width		1.2 – 1.6 mm	0.047 – 0.063 in.	
Valve guide bushing	Inside diameter		6.010 – 6.030 mm	0.2366 – 0.2374 in.	
	Outside diameter (for repair part)	STD	11.048 – 11.059 mm	0.4350 – 0.4354 in.	
		O/S 0.05	11.098 – 11.109 mm	0.4369 – 0.4374 in.	

Specifications (Cont'd)

Valve	Valve overall length	STD	Intake	91.45 mm	3.6004 in.
			Exhaust	91.90 mm	3.6181 in.
		Limit	Intake	90.95 mm	3.5807 in.
	Valve face angle		Exhaust	91.40 mm	3.5984 in.
	Stem diameter			44.5°	
			Intake	5.970 – 5.985 mm	0.2350 – 0.2356 in.
			Exhaust	5.965 – 5.980 mm	0.2348 – 0.2354 in.
	Stem oil clearance	STD	Intake	0.025 – 0.060 mm	0.0010 – 0.0024 in.
			Exhaust	0.030 – 0.065 mm	0.0012 – 0.0026 in.
		Limit	Intake	0.08 mm	0.0031 in.
	Margin thickness		Exhaust	0.10 mm	0.0039 in.
		STD		0.8 – 1.2 mm	0.031 – 0.047 in.
			Limit		
				0.5 mm	0.020 in.
Valve spring	Squareness		Limit	2.0 mm 0.079 in.	
	Free length			43.8 mm 1.774 in.	
	Installed tension at 34.7 mm (1.366 in.)			143 – 155 N (14.6 – 15.8 kgf, 32.2 – 34.8 lbf)	
Valve lifter	Lifter diameter			27.975 – 27.985 mm	1.1014 – 1.1018 in.
	Lifter bore diameter			28.005 – 28.026 mm	1.1026 – 1.1034 in.
	Oil clearance	STD		0.020 – 0.051 mm	0.0008 – 0.0020 in.
		Limit		0.07 mm	0.0028 in.
Manifold	Warpage	Limit	Intake	0.20 mm	0.0079 in.
			Exhaust	0.30 mm	0.0118 in.
Camshaft and gear	Thrust clearance	STD	Intake	0.030 – 0.085 mm	0.0012 – 0.0033 in.
			Exhaust	0.035 – 0.090 mm	0.0014 – 0.0035 in.
		Limit		0.11 mm	0.0043 in.
	Journal oil clearance	STD		0.035 – 0.072 mm	0.0014 – 0.0028 in.
		Limit		0.10 mm	0.0039 in.
	Journal diameter	Exhaust	No.1	24.949 – 24.965 mm	0.9822 – 0.9829 in.
		Others		22.949 – 22.965 mm	0.9035 – 0.9041 in.
	Circle runout		Limit	0.04 mm	0.0016 in.
	Cam lobe height	STD	Intake	35.210 – 35.310 mm	1.3862 – 1.3902 in.
			Exhaust	34.910 – 35.010 mm	1.3744 – 1.3783 in.
		Limit	Intake	34.81 mm	1.3705 in.
			Exhaust	34.51 mm	1.3587 in.
	Camshaft gear backlash	STD		0.020 – 0.200 mm	0.0008 – 0.0079 in.
		Limit		0.30 mm	0.0188 in.
	Camshaft gear spring end free distance			17.0 – 17.6 mm	0.669 – 0.693 in.
Cylinder block	Cylinder head surface warpage		Limit	0.05 mm	0.0020 in.
	Cylinder bore diameter	STD	Mark 1	81.000 – 81.010 mm	3.1890 – 3.1894 in.
			Mark 2	81.010 – 81.020 mm	3.1894 – 3.1898 in.
			Mark 3	81.020 – 81.030 mm	3.1898 – 3.1902 in.
		Limit	STD	81.23 mm	3.1980 in.
			O/S 0.50	81.73 mm	3.2177 in.

Specifications (Cont'd)

Piston and piston ring	Piston diameter	STD	Mark 1	80.930 – 80.940 mm	3.1862 – 3.1866 in.
			Mark 2	80.940 – 80.950 mm	3.1866 – 3.1870 in.
			Mark 3	80.950 – 80.960 mm	3.1870 – 3.1874 in.
		O/S 0.50		81.430 – 81.460 mm	3.2059 – 3.2071 in.
	Piston oil clearance	STD		0.060 – 0.080 mm	0.0024 – 0.0031 in.
		Limit		0.10 mm	0.0039 in.
	Piston ring groove clearance		No.1	0.040 – 0.081 mm	0.0016 – 0.0032 in.
			No.2	0.030 – 0.070 mm	0.0012 – 0.0028 in.
	Piston ring end gap	STD	No.1	0.250 – 0.450 mm	0.0098 – 0.0177 in.
			No.2	0.150 – 0.400 mm	0.0059 – 0.0157 in.
			Oil	0.100 – 0.700 mm	0.0039 – 0.0276 in.
		Limit	No.1	1.05 mm	0.0413 in.
			No.2	1.00 mm	0.0394 in.
			Oil	1.30 mm	0.0512 in.
Connecting rod	Thrust clearance	STD		0.150 – 0.250 mm	0.0059 – 0.0098 in.
		Limit		0.30 mm	0.0118 in.
	Connecting rod bearing center wall thickness				
		STD	Mark 1	1.486 – 1.490 mm	0.0585 – 0.0587 in.
			Mark 2	1.490 – 1.494 mm	0.0587 – 0.0588 in.
			Mark 3	1.494 – 1.498 mm	0.0588 – 0.0590 in.
	Connecting rod oil clearance	STD	STD	0.020 – 0.051 mm	0.0008 – 0.0020 in.
			U/S 0.25	0.019–0.065 mm	0.0007 – 0.0026 in.
		Limit		0.08 mm	0.0031 in.
	Rod bending Limit per 100 mm (3.94 in.)			0.05 mm	0.0020 in.
	Rod twist Limit per 100 mm (3.94 in.)			0.05 mm	0.0020 in.
Crankshaft	Thrust clearance		Limit	0.020 – 0.220 mm	0.0008 – 0.0087 in.
				0.30 mm	0.0118 in.
	Thrust washer thickness			2.440 – 2.490 mm	0.0961 – 0.0980 in.
	Main journal oil clearance	STD	STD	0.015–0.033 mm	0.0006 – 0.0013 in.
			U/S 0.25	0.018 – 0.056 mm	0.0007 – 0.0022 in.
		Limit		0.10 mm	0.0039 in.
	Main journal diameter		STD	47.982 – 48.000 mm	1.8891 – 1.8898 in.
			U/S 0.25	47.745 – 47.755 mm	1.8797 – 1.8881 in.
	Main bearing center wall thickness				
		STD	Mark 1	2.002 – 2.005 mm	0.0788 – 0.0789 in.
			Mark 2	2.005 – 2.008 mm	0.0789 – 0.0791 in.
			Mark 3	2.008 – 2.011 mm	0.0791 – 0.0792 in.
			Mark 4	2.011 – 2.014 mm	0.0792 – 0.0793 in.
			Mark 5	2.014 – 2.017 mm	0.0793 – 0.0794 in.
	Crank pin diameter	STD		39.985 – 40.000 mm	1.5742 – 1.5748 in.
			U/S 0.25	39.745 – 39.755 mm	1.5648 – 1.5652 in.
	Circle runout		Limit	0.06 mm	0.0024 in.
	Main journal taper and out-of-round		Limit	0.02 mm	0.0008 in.
	Crank pin taper and out-of-round		Limit	0.02 mm	0.0008 in.

Torque Specifications

Part tightened	N·m	kgf-cm	ft-lbf
Crankshaft pulley x Crankshaft	118	1,200	87
Camshaft timing pulley x Camshaft	59	600	43
Idler pulley x Cylinder block	37	375	27
Cylinder head x Cylinder block	60	610	44
Camshaft bearing cap x Cylinder head	13	130	9
PS drive belt adjusting strut x Cylinder head	39	400	29
Engine hanger x Cylinder head	27	280	20
Fan belt adjusting bar x Cylinder head	20	200	14
Cylinder head cover x Cylinder head	7.8	80	69 in.-lbf
Intake manifold x Cylinder head	19	195	14
Intake manifold stay x Intake manifold	19	195	14
Intake manifold stay x Cylinder block	39	400	29
EGR valve x Intake manifold	13	130	9
ACV x Intake manifold	13	130	9
Water inlet housing x Cylinder head	20	200	14
Water outlet x Cylinder head	15	150	11
Exhaust manifold x Cylinder head	25	250	18
Exhaust manifold stay x Exhaust manifold	39	400	29
Exhaust manifold stay x Cylinder block	39	400	29
Main bearing cap x Cylinder block	60	610—	44
Connecting rod cap x Connecting rod	49	500	36
Rear oil seal retainer x Cylinder block	9.3	95	82 in.-lbf
PS pump bracket x Cylinder block	19	195	14
RH engine mounting bracket x Cylinder block	51	525	38
Alternator bracket x Cylinder block	39	400	29
Rear end plate x Cylinder block	9.3	95	82 in.-lbf
Flywheel (M /T) x Crankshaft	78	800	58
Drive plate (A/T) x Crankshaft	64	650	47
LH engine mounting bracket x Transaxle	52	530	38
LH engine mounting insulator x LH mounting bracket	48	490	35
LH engine mounting insulator x Body	87	890	64
R H engine mounting insulator x R H mounting bracket Nut	52	530	38
Bolt	64	650	47
RH engine mounting insulator x Body	87	890	64
R H engine mounting stay x R H mounting insulator	42	430	31
RH engine mounting stay x PS drive belt adjusting strut	42	430	31
LH engine mounting stay x Transaxle	21	210	15
LH engine mounting stay x LH mounting insulator	21	210	15
Front engine mounting bracket x Transaxle	77	790	57
Rear engine mounting bracket x Transaxle	77	790	57
Engine mounting center member x Body	52	530	38
Engine mounting center member x Front mounting insulator	64	650	47
Engine mounting center member x Rear mounting insulator	64	650	47
Front engine mounting bracket x Front mounting insulator	87	890	64
Rear engine mounting bracket x Rear mounting insulator	87	890	64

Torque Specifications (Cont'd)

Part tightened	N-m	kgf-cm	ft-lbf
PS pump x Bracket	39	400	29
A/C compressor x Bracket	25	250	18
Suspension lower crossmember x Body	152	1,550	112

ENGINE MECHANICAL (3S-GTE) Specifications

Idle speed				800 ± 50 rpm
Intake manifold vacuum	at idle speed			60 kPa 450 mmHg 17.7 in.Hg
Compression	at 250 rpm	STD Limit		1,128 kPa (11.5 kgf/cm ² , 164 psi) or more 883 kPa 9.0 kgf/cm ² 128 psi 98 kPa (1.0 kgf/cm ² , 14 psi) or less
	Difference of pressure between each cylinder			
Timing belt tensioner	Protrusion			8.5 – 9.5 mm 0.335 – 0.374 in.
Cylinder head	warpage	Cylinder block side Intake manifold side Exhaust manifold side	Limit Limit Limit	0.20 mm 0.0079 in. 0.20 mm 0.0079 in. 0.30 mm 0.0118 in.
	Valve seat	Refacing angle Contacting angle Contacting width		30°, 45°, 75° 45° 1.0 – 1.4 mm 0.039 – 0.055 in.
Valve guide bushing	Inside diameter			6.000 – 6.018 mm 0.2362 – 0.2369 in.
	Outside diameter (for repair part)	STD O/S 0.05		11.030 – 11.041 mm 0.4343 – 0.4347 in. 11.080 – 11.091 mm 0.4362 – 0.4367 in.
Valve	Valve overall length	STD	Intake Exhaust	100.50 mm 3.9567 in. 99.55 mm 3.9193 in.
		Limit	Intake Exhaust	99.80 mm 3.9291 in. 98.85 mm 3.8917 in.
	Valve face angle			44.5°
	Stem diameter		Intake Exhaust	5.960 – 5.975 mm 0.2346 – 0.2352 in. 5.955 – 5.970 mm 0.2344 – 0.2350 in.
	Stem oil clearance	STD	Intake Exhaust	0.025 – 0.058 mm 0.0010 – 0.0023 in. 0.030 – 0.063 mm 0.0012 – 0.0025 in.
		Limit	Intake Exhaust	0.08 mm 0.0031 in. 0.10 mm 0.0039 in.
	Margin thickness	STD Limit		0.8 – 1.2 mm 0.031 – 0.047 in. 0.5 mm 0.020 in.
Valve spring	Squareness	Limit		2.0 mm 0.079 in.
	Free length			44.43 mm 1.7492 in.
	Installed tension at 34.4 mm (1.354 in.)			201 – 236 N (20.5 – 24.1 kgf, 45.2 – 53.1 lbf)

Specifications (Cont'd)

Valve lifter	Lifter diameter		30.975 – 30.985 mm 1.2195 – 1.2199 in.
	Lifter bore diameter		31.000 – 31.021 mm 1.2205 – 1.2213 in.
	Oil clearance	STD Limit	0.015 – 0.046 mm 0.0006 – 0.0018 in. 0.07 mm 0.0028 in.
Manifold	Warpage	Limit	0.20 mm 0.0079 in.
Camshaft	Thrust clearance	STD Limit	0.120 – 0.240 mm 0.0047 – 0.0094 in. 0.30 mm 0.0118 in.
	Journal oil clearance	STD Limit	0.025 – 0.062 mm 0.0010 – 0.0024 in. 0.08 mm 0.0031 in.
	Journal diameter		26.959 – 26.975 mm 1.0614 – 1.0620 in.
	Circle runout	Limit	0.06 mm 0.0024 in.
	Cam lobe height	STD Intake Exhaust Limit Intake Exhaust	41.010–41.110mm 1.6146–1.6185 in. 41.090 – 41.190 mm 1.6177 –1.6217 in. 39.90 mm 1.5709 in. 39.98 mm 1.5740 in.
T-VIS valve	Warpage	Limit	0.20 mm 0.0079 in.
Cylinder block	Cylinder head surface warpage	Limit	0.05 mm 0.0020 in.
	Cylinder bore diameter	Mark 1	86.000 – 86.010 mm 3.3858 – 3.3862 in.
		Mark 2	86.010 – 86.020 mm 3.3862 – 3.3866 in.
		Mark 3	86.020 – 86.030 mm 3.3866 – 3.3870 in.
		Limit	86.23 mm 3.3949 in.
Piston and piston ring	Piston diameter	Mark 1	85.920 – 85.930 mm 3.3827 – 3.3831 in.
		Mark 2	85.930 – 85.940 mm 3.3831 – 3.3835 in.
		Mark 3	85.940 – 85.950 mm 3.3835 – 3.3839 in.
	Piston oil clearance	STD Limit	0.070 – 0.090 mm 0.0028 – 0.0035 in. 0.110 mm 0.0043 in.
	Piston ring groove clearance	No.1 No.2	0.040 – 0.080 mm 0.0016 – 0.0031 in. 0.030 – 0.070 mm 0.0012 – 0.0028 in.
	Piston ring end gap	STD No.1 No.2 Oil Limit No.1 No.2 Oil	0.330 – 0.550 mm 0.0130 – 0.0217 in. 0.450 – 0.670 mm 0.0177 – 0.0264 in. 0.200 – 0.600 mm 0.0079 – 0.0236 in. 0.85 mm 0.0335 in. 0.97 mm 0.0382 in. 0.90 mm 0.0354 in.
Connecting rod	Thrust clearance	STD	0.160 – 0.312 mm 0.0063 – 0.0123 in.
	Limit		0.35 mm 0.0138 in.
	Connecting rod bearing center wall thickness		
		STD Mark 1	1.484 – 1.488 mm 0.0584 – 0.0586 in.
		Mark 2	1.488 – 1.492 mm 0.0586 – 0.0587 in.
		Mark	1.492 – 1.496 mm 0.0587 – 0.0589 in.
	Connecting rod oil clearance	STD STD	0.024 – 0.055 mm 0.0009 – 0.0022 in.
		U/S 0.25	0.023 – 0.069 mm 0.0009 – 0.0027 in.
	Limit		0.08 mm 0.0031 in.
	Rod bending Limit per 100 mm (3.94 in.)		0.05 mm 0.0020 in.
	Rod twist Limit per 100 mm (3.94 in.)		0.15 mm 0.0059 in.

Specifications (Cont'd)

Connecting rod (cont'd)	Bushings inside diameter		22.005 – 22.017 mm	0.8663 – 0.8668 in.
	Piston pin diameter		21.997 – 22.009 mm	0.8660 – 0.8665 in.
	Piston pin oil clearance	STD	0.005 – 0.011 mm	0.0002 – 0.0004 in.
		Limit	0.05 mm	0.0020 in.
Crankshaft	Thrust clearance	STD	0.020 – 0.220 mm	0.0008 – 0.0087 in.
		Limit	0.30 mm	0.0118 in.
	Thrust washer thickness		2.440 – 2.490 mm	0.0961 – 0.0980 in.
	Main journal oil clearance			
	STD	No-3 STD	0.025 – 0.044 mm	0.0010 – 0.0017 in.
		U/S 0.25	0.021 – 0.061 mm	0.0008 – 0.0024 in.
		Others STD	0.015 – 0.034 mm	0.0006 – 0.0013 in.
		U/S 0.25	0.029 – 0.069 mm	0.0011 – 0.0027 in.
	Limit		0.08 mm	0.0031 in.
	Main journal diameter	STD	54.988 – 55.003 mm	2.1653 – 2.1655 in.
		U/S 0.25	54.745 – 54.755 mm	2.1553 – 2.1557 in.
	Main bearing center wall thickness			
	STD No.3	Mark 1	1.992 – 1.995 mm	0.0784 – 0.0785 in.
		Mark 2	1.995 – 1.998 mm	0.0785 – 0.0787 in.
		Mark 3	1.998 – 2.001 mm	0.0787 – 0.0788 in.
		Mark 4	2.001 – 2.004 mm	0.0788 – 0.0789 in.
		Mark 5	2.004 – 2.007 mm	0.0789 – 0.0790 in.
	Others	Mark 1	1.997 – 2.000 mm	0.0786 – 0.0787 in.
		Mark 2	2.000 – 2.003 mm	0.0787 – 0.0789 in.
		Mark 3	2.003 – 2.006 mm	0.0789 – 0.0790 in.
		Mark 4	2.006 – 2.009 mm	0.0790 – 0.0791 in.
		Mark 5	2.009 – 2.012 mm	0.0791 – 0.0792 in.
	Crank pin diameter	STD	47.985 – 48.000 mm	1.8892 – 1.8898 in.
		U/S 0.25	47.745 – 47.755 mm	1.8797 – 1.8801 in.
	Circle runout	Limit	0.06 mm	0.0024 in.
		Limit	0.02 mm	0.0008 in.
	Main journal taper and out-of-round	Limit	0.02 mm	0.0008 in.
	Crank pin taper and out-of-round			

Torque Specifications

Part tightened		N·m	kgf-cm	ft-lbf
Oil pump pulley x Oil pump drive shaft		35	355	26
No.2 idler pulley x Cylinder block		43	440	32
No.1 idler pulley bracket x Cylinder head		52	530	38
Crankshaft pulley x Crankshaft		108	1,100	80
Camshaft timing pulley x Camshaft		59	600	43
Timing belt tensioner x Cylinder head	For SST	41	420	30
		21	210	15
Cylinder head x Cylinder block	1st	49	500	36
	2nd	Turn 90°		
Camshaft bearing cap x Cylinder head		19	190	14
No.3 timing belt cover x Cylinder head		8.8	90	78 in.-lbf

Torque Specifications (Cont'd)

Part tightened		N·m	kgf-cm	ft-lbf
Cylinder head cover x Cylinder head		2.5	25	21 in-lbf
Intake manifold x Cylinder head		19	195	14
Intake manifold stay x Intake manifold		25	260	19
Intake manifold stay x Cylinder block		25	260	19
Water by-pass pipe x Water pump cover		7.8	80	69 in-lbf
Water outlet x Cylinder head		39	400	29
EGR valve x Intake manifold		19	195	14
EGR pipe x Cylinder head		25	260	19
LH engine hanger x Cylinder head	12 mm head bolt	19	195	14
	14 mm head bolt	39	400	29
Exhaust manifold x Cylinder head		52	530	38
Catalytic converter x Turbine outlet elbow		29	300	22
Catalytic converter stay x Catalytic converter		59	600	43
No.1 alternator bracket x Cylinder head		39	400	29
Main bearing cap x Cylinder block		59	600	43
Connecting rod cap x Connecting rod		67	680	49
Rear oil seal retainer x Cylinder block		9.3	95	82 in-lbf
Knock sensor x Cylinder block		44	450	33
RH engine mounting bracket x Cylinder block		52	530	38
PS pump bracket x Cylinder block		43	440	32
Rear end plate x Cylinder block		9.3	95	82 in-lbf
Flywheel x Crankshaft		108	1,100	80
LH engine mounting bracket x Transaxle		52	530	38
LH engine mounting insulator x LH mounting bracket		63	650	47
LH engine mounting insulator x Body		87	890	64
RH engine mounting insulator x RH mounting bracket		52	530	38
RH engine mounting insulator x Body		87	890	64
RH engine mounting stay x RH mounting insulator		73	740	54
RH engine mounting stay x No.1 alternator bracket		73	740	54
LH engine mounting stay x LH mounting insulator		21	210	15
LH engine mounting stay x Transaxle		21	210	15
Front engine mounting bracket x Transaxle		77	790	57
Rear engine mounting bracket x Transaxle		77	790	57
Engine mounting center member x Body		52	530	38
Engine mounting center member x Front mounting insulator		73	740	54
Engine mounting center member x Rear mounting insulator		73	740	54
Front engine mounting insulator x Front mounting insulator		87	890	64
Rear engine mounting insulator x Rear mounting insulator		87	890	64
PS pump x PS pump bracket	Adjusting bolt	39	400	29
	Others	43	440	32
A/C compressor x Cylinder block		27	280	20
Suspension lower crossmember x Body		152	1,550	112
Transaxle oil cooler tube x Oil cooler hose		34	350	25
Suspension upper brace x Body	Bolt	21	210	15
	Nut	64	650	47

ENGINE MECHANICAL (5S-FE)

Specifications

Idle speed		USA CANADA	700 ± 50 rpm 750 ± 50 rpm
Intake manifold vacuum	at idle speed		60 kPa 450 mmHg 17.7 in.Hg
Compression pressure	at 250 rpm Difference of pressure between each cylinder	STD Limit	1,226 kPa (12.5 kgf/cm ² , 178 psi) or more 981 kPa 10.0 kgf/cm ² 142 psi 98 kPa (1.0 kgf/cm ² , 14 psi) or less
Idle pulley tension spring	Free length at 51.9 mm (2.043 in.) Installed load		46.0 mm 1.811 in. 32–37 N 3.25–3.75 kgf 7.2–8.3 lbf
Cylinder head	Warpage Cylinder block side Manifold side Valve seat Refacing angle Contacting angle Contacting width	Limit Limit	0.05 mm 0.020 in. 0.08 mm 0.031 in. 30°, 45°, 75° 45° 1.0 – 1.4 mm 0.039 – 0.055 in.
Valve guide bushing	Inside diameter Outside diameter (for repair part)	STD O/S 0.05	6.010 – 6.030 mm 0.2366 – 0.2374 in. 11.033 – 11.044 mm 0.4344 – 0.4348 in. 11.083 – 11.094 mm 0.4363 – 0.4368 in.
Valve	Valve overall length Valve face angle Stem diameter Stem oil clearance Margin thickness	STD Limit Intake Exhaust Intake Exhaust Intake Exhaust STD Limit STD Limit	97.60 mm 3.8425 in. 98.45 mm 3.8760 in. 97.1 mm 3.823 in. 98.0 mm 3.858 in. 44.5° 5.970 – 5.985 mm 0.2350 – 0.2356 in. 5.965 – 5.980 mm 0.2348 – 0.2354 in. 0.025 – 0.060 mm 0.0010 – 0.0024 in. 0.030 – 0.065 mm 0.0012 – 0.0026 in. 0.08 mm 0.0031 in. 0.10 mm 0.0039 in. 0.8 – 1.2 mm 0.031 – 0.047 in. 0.5 mm 0.020 in.
Valve spring	Squareness Free length Installed tension at 34.7 mm (1.366 in.)	Limit	2.0 mm 0.079 in. 41.96 – 41.99 mm 1.6520 1.6520 – 1.6531 in. 164 – 189 N 0 6.7 – 19.3 kgf, 36.8 – 42.5 lbf)
Valve lifter	Lifter diameter Lifter bore diameter Oil clearance	STD Limit	30.966 – 30.976 mm 1.2191 – 1.2195 in. 31.000 – 31.018 mm 1.2205 – 1.2213 in. 0.024 – 0.052 mm 0.0009 – 0.0020 in. 0.07 mm 0.0028 in.
Manifold	Warpage	Limit	0.30 mm 0.0118 in.

Specifications (Cont'd)

Camshaft	Thrust clearance	STD	Intake	0.045 – 0.100 mm 0.0018 – 0.0039 in.
			Exhaust	0.030 – 0.085 mm 0.0012 – 0.0033 in.
		Limit	Intake	0.12 mm
			Exhaust	0.0047 in.
	Journal oil clearance	STD		0.10 mm 0.0039 in.
		Limit		0.025 – 0.062 mm 0.0010 – 0.0024 in.
	Journal diameter			0.10 mm 0.0039 in.
	Circle runout Limit			26.959 – 26.975 mm 1.0614 – 1.0620 in.
	Cam lobe height STD Intake			0.04 mm 0.0016 in.
	Exhaust			42.010 – 42.110 mm 1.6539 – 1.6579 in.
	Limit Intake			40.060 – 40.160 mm 1.5772 – 1.5811 in.
	Exhaust			41.90 mm 1.6496 in.
Cylinder block	Camshaft gear backlash STD			39.95 mm 1.5728 in.
	Limit			0.020 – 0.200 mm 0.0008 – 0.0079 in.
	Camshaft gear spring end free distance			0.30 mm 0.0188 in.
				22.5 – 22.9 mm 0.886 – 0.902 in.
	Cylinder head surface warpage	Limit		0.05 mm 0.0020 in.
	Cylinder bore diameter STD	Mark 1		87.000 – 87.010 mm 3.4252 – 3.4256 in.
Piston and piston ring		Mark 2		87.010 – 87.020 mm 3.4256 – 3.4260 in.
		Mark 3		87.020 – 87.030 mm 3.4260 – 3.4264 in.
		Limit		87.23 mm 3.4342 in.
		O/S 0.50		87.73 mm 3.4350 in.
	Piston diameter	STD	Mark 1	86.850 – 86.860 mm 3.4193 – 3.4197 in.
			Mark 2	86.860 – 86.870 mm 3.4197 – 3.4201 in.
			Mark 3	86.870 – 86.880 mm 3.4201 – 3.4205 in.
		O/S 0.50		87.350 – 87.380 mm 3.4390 – 3.4402 in.
	Piston oil clearance	STD		0.140 – 0.160 mm 0.0055 – 0.0063 in.
		Limit		0.18 mm 0.0071 in.
	Piston ring groove clearance	No.1		0.040 – 0.080 mm 0.0016 – 0.0031 in.
		No.2		0.030 – 0.070 mm 0.0012 – 0.0028 in.
Connecting rod	Piston ring end gap	STD	No.1	0.270–0.500 mm 0.0106–0.0197 in.
			No.2	0.350 – 0.600 mm 0.0138 – 0.0234 in.
			Oil	0.200 – 0.550 mm 0.0079 – 0.0217 in.
		Limit	No.1	1.10 mm 0.0433 in.
			No.2	1.20 mm 0.0472 in.
			Oil	1.15 mm 0.0453 in.
	Thrust clearance STD			0.160 – 0.312 mm 0.0063 – 0.0123 in.
	Limit			0.35 mm 0.0138 in.
	Connecting rod bearing center wall thickness			
		STD	Mark 1	1.484 – 1.488 mm 0.0584 – 0.0586 in.
			Mark 2	1.488 – 1.492 mm 0.0586 – 0.0587 in.
			Mark 3	1.492 – 1.496 mm 0.0587 – 0.0589 in.
	Connecting rod oil clearance			
		STD	STD	0.024 – 0.055 mm 0.0009 – 0.0022 in.
			U/S 0.25	0.023 – 0.069 mm 0.0009 – 0.0027 in.
		Limit		0.08 mm 0.0031 in.
	Rod bending Limit per 100 mm (3.94 in.)			0.05 mm 0.0020 in.
	Rod twist Limit per 100 mm (3.94 in.)			0.15 mm 0.0059 in.
	Bushing inside diameter			22.005 – 22.017 mm 0.8663 – 0.8668 in.

Specifications (Cont'd)

Connecting rod (cont'd)	Piston pin diameter			21.997 – 22.009 mm	0.8660 – 0.8665 in.
	Piston pin oil clearance			0.005 – 0.011 mm	0.0002 – 0.0004 in.
	Limit			0.05 mm	0.0020 in.
	Connecting rod bolt outside diameter				
	STD			7.860 – 8.000 mm	0.3094 – 0.3150 in.
	Limit			7.60 mm	0.2992 in.
Crankshaft	Thrust clearance			0.020 – 0.220 mm	0.0008 –
	Limit			0.30 mm	0.0087 in.
	Thrust washer thickness			2.440 – 2.490 mm	0.0118 in.
	Main journal oil clearance				0.0961 – 0.0980 in.
	STD	N o. 3	STD	0.025 – 0.044 mm	0.0010 – 0.0017 in.
			U/S 0.25	0.027 – 0.067 mm	0.0011 – 0.0026 in.
		Others	STD	0.015–0.034 mm	0.0006 – 0.0013 in.
			U/S 0.25	0.019 – 0.059 mm	0.0007 – 0.0023 in.
	Limit			0.08 mm	0.0031 in.
	Main journal diameter			54.988 – 55.003 mm	2.1653 – 2.1655 in.
			U/S 0.25	54.745 – 54.755 mm	2.1553 – 2.1557 in.
	Main bearing center wall thickness				
	STD	No.3	Mark 1	1.992 – 1.995 mm	0.0784 – 0.0785 in.
			Mark 2	1.995 – 1.998 mm	0.0785 – 0.0787 in.
			Mark 3	1.998 – 2.001 m m	0.0787 – 0.0788 in.
			Mark 4	2.001 – 2.004 m m	0.0788 – 0.0789 in.
			Mark 5	2.004 – 2.007 m m	0.0789 – 0.0790 in.
		Others	Mark 1	1.997 – 2.000 mm	0.0786 – 0.0787 in.
			Mark 2	2.000 – 2.003 m m	0.0787 – 0.0789 in.
			Mark 3	2.003 – 2.006 mm	0.0789 – 0.0790 in.
			Mark 4	2.006 – 2.009 mm	0.0790 – 0.0791 in.
			Mark 5	2.009–2.012 mm	0.0791 – 0.0792 in.
	Crank pin diameter			51.985 – 52.000 mm	2.0466 – 2.0472 in.
			U/S 0.25	51.745 – 51.755 mm	2.0372 – 2.0376 in.
	Circle runout			0.06 mm	0.0024 in.
	Main journal taper and out-of-round			0.02 mm	0.0008 in.
	Crank pin taper and out-of-round			0.02 mm	0.0008 in.

Torque Specifications

Part tightened		N·m	kgf-cm	ft-lbf
Oil pump pulley x Oil pump drive shaft		28	290	21
No.2 idler pulley x Cylinder block		42	425	31
Crankshaft pulley x Crankshaft		108	1,100	80
Camshaft timing pulley x Camshaft		54	550	40
		37	380	27
No.1 idler pulley x Cylinder head		42	425	31
Cylinder head x Cylinder block		49	500	36
		Turn 90°		
Spark plug tube x Cylinder head		39	400	29
Camshaft bearing cap x Cylinder head		19	190	14

Torque Specifications (Cont'd)

Part tightened		N·m	kgf-cm	ft-lbf
Cylinder head cover x Cylinder head		23	230	17
Alternator bracket x Cylinder head		42	425	31
Engine hanger x Cylinder head		25	250	18
No.3 timing belt cover x Cylinder head		7.8	80	fig in.-lbf
Intake manifold x Cylinder head		19	195	14
Intake manifold stay x Intake manifold		19	195	14
Intake manifold stay x Cylinder block		42	425	31
EGR valve x Intake manifold		13	130	9
EGR pipe x Cylinder head		59	600	43
Water by-pass pipe x Water pump cover		9.3	95	82 in.-lbf
Water outlet x Cylinder head		15	150	11
Catalytic converter x Exhaust manifold		29	300	22
Exhaust manifold x Cylinder head		49	500	36
Catalytic converter stay x Catalytic converter		42	425	31
Catalytic converter stay x Cylinder block		42	425	31
Main bearing cap x Cylinder block		59	600	43
Connecting rod cap x Connecting rod	1st	25	250	18
	2nd	Turn 90°		
Rear oil seal retainer x Cylinder block		9.3	95	82 in.-lbf
Knock sensor x Cylinder block		37	380	27
Rear end plate x Cylinder block		9.3	95	82 in.-lbf
Flywheel x Crankshaft (M /T)		88	900	65
Drive plate x Crankshaft (A/T)		83	850	61
R H engine mounting bracket x Cylinder block		52	530	38
PS pump bracket x Cylinder head		43	440	32
LH engine mounting bracket x Transaxle		52	530	38
LH engine mounting insulator x Body		87	890	64
LH engine mounting insulator x LH mounting bracket		63	650	47
RH engine mounting insulator x Body		87	890	64
RH engine mounting insulator x RH mounting bracket		52	530	38
R H engine mounting stay x R H mounting insulator		73	740	54
RH engine mounting stay x Alternator bracket		73	740	54
LH engine mounting stay x Transaxle		21	210	15
LH engine mounting stay x LH mounting insulator		21	210	15
Front engine mounting bracket x Transaxle		77	790	57
Rear engine mounting bracket x Transaxle		77	790	57
Engine mounting center member x Body		52	530	38
Engine mounting center member x Front mounting insulator		73	740	54
Engine mounting center member x Rear mounting insulator		73	740	54
Front engine mounting bracket x Front mounting bracket		87	890	64
Rear engine mounting bracket x Rear mounting insulator		87	890	64
PS pump x PS pump bracket	Adjusting bolt	39	400	29
	Others	43	440	32
A/C compressor x Cylinder block		27	280	20
Suspension lower crossmember x Body		152	1,550	112
Suspension upper brace x Body	Bolt	21	210	15
	Nut	64	650	47

EXHAUST SYSTEM

Part tightened	N-m	kgf-cm	ft-lbf
Front exhaust pipe x Exhaust manifold (4A–FE)	62	630	46
Front exhaust pipe x Catalytic converter (4A–FE)	43	440	32
Front exhaust pipe x Catalytic converter (3S–GTE and 5S–FE)	62	630	46
Center exhaust pipe x Catalytic converter (4A–FE)	43	440	32
Front exhaust pipe x Center exhaust pipe (3S–GTE and 5S–FE)	43	440	32
Center exhaust pipe x Tailpipe (4A–FE and 3S–GTE)	43	440	32
Center exhaust pipe x Tailpipe (5S–FE)	21	210	15

TURBOCHARGER SYSTEM

Specifications

Turbocharger	Turbocharging pressure	49 – 81 kPa (0.50 – 0.83 kgf/cm ² , 7.1 – 11.8 psi)
	Impeller wheel axial play	0.13 mm (0.0051 in.) or less
	Impeller wheel radial play	0.18 mm (0.0071 in.) or less

Torque Specifications

Part tightened	N-m	kgf-cm	ft-lbf
Turbine outlet elbow x Turbocharger	64	650	47
Side bearing housing plate x Turbocharger	11	120	9
Turbo water pipe x Turbocharger	11	120	9
Turbocharger x Exhaust manifold	64	650	47
Turbo oil pipe x Turbocharger	17	175	13
Turbo oil pipe x Cylinder block	43	440	32
	Bolt	51	38
	Union bolt	69	51
Turbocharger stay x Turbocharger	59	600	43
Turbocharger stay x Cylinder block	44	450	33
Oxygen sensor x Turbine outlet elbow			

MFI AND SFI SYSTEMS (4A–FE)

Specifications

Fuel pressure regulator	Fuel pressure at no vacuum	2655 – 304 kPa (2.7 – 3.1 kgf/cm ² , 38 – 44 psi)
Injector	Resistance Injection volume Difference between each injector Fuel leakage	Approx. 13.8 kΩ 40 – 50 cm ³ (2.4 – 3.1 cu in.)/15 sec. 5 cm ³ (0.31 cu in.) or less One drop or less per minute
Throttle body	Throttle body fully closed angle	6°

Specifications (Cont'd)

Throttle position sensor	Throttle opening angle (from vertical)	Clearance between stop screw and lever		IDL – E2	PSW – E2
	—	0.60 mm 0.024 in.		Continuity	No continuity
	—	0.80 m m 0.032 in.		No continuity	No continuity
	Throttle valve fully open	—		No continuity	Continuity
	63°	—		No continuity	No continuity
	69°	—		No continuity	Continuity
	7.5° or less	—		Continuity	No continuity
Dash pot	Setting speed	M/T	1,800 rpm		
		Afr	2,200 rpm		
ACV valve	Resistance		27–33Ω		
EGR VSV	Resistance		33–39Ω		
Water temp. sensor–	Resistance	at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (104°F) at 60°C (140°F) at 80°C (176°F)	10–20 kΩ 4–7 kΩ 2–3 kΩ 0.9 – 1.3 kΩ 0.4 – 0.7 kΩ 0.2 – 0.4 kΩ		
Intake air temp. sensor	Resistance	at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (104°F) at 60°C (140°F) at 80°C (176°F)	10–20 kΩ 4–7 kΩ 2–3 kΩ 0.9 – 1.3 kΩ 0.4 – 0.7 kΩ 0.2 – 0.4 kΩ		
EGR gas temp. sensor (CALIF. only)	Resistance	at 500C (112°F) at 1001C 1212°F) at 1501C (302°F)	69 – 89 Ω 11–15kΩ 2–4 kΩ		
Oxygen sensor heater (Ex. CALIF.)	Resistance		5.1 –6.3Ω		
ECU	HINT: • Perform all voltage and resistance measurements with the ECU connected. • Verity that the battery voltage is 11 V or above with the ignition switch ON.				
	Voltage				
	Terminals	Condition			STD voltage (V)
	+B _ E1 +B1	!G SW ON			10–14
	BATT – E1				10–14
	IDL – E2	IG SW ON	Throttle valve open		10–14
	PSW – E2		Throttle valve fully closed		10–14
	PIM – E2	IG SW ON			3.3–3.9
	VCC – E2				4.5–5.5

Specifications (Cont'd)

ECU (cont'd)	Voltage (cont'd)			
	Terminals	Condition	STD voltage (V)	
	No.10 _ E01 No. 20 E02	IG SW ON	10-14	
	THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1-3
	THW – E2		Coolant temp. 80°C (176°F)	0.1 – 1.0
	STA – E1	Cranking		6-14
	IGT – E1	Cranking or idling		0.7-1.0
	V11 – E1	No trouble ("CHECK" engine warning light off) and engine running		10-14
	A/C – E1	IG SW ON	Air conditioning ON	8-14
	ACT – E1		Air conditioning ON	4-6
	T – E1		Check connector TE1 – E1 not connected	10-14
			Check connector TE1 – E1 connected	0.5 or less
	NSW – E1		Neutral start switch P or N range	0-2
			Ex. neutral start switch P or N range	6-14
	V-ISC – E1	Cranking for ten seconds after starting		10-14
	Resistance			
	Terminals	Condition		STD resistance (Ω)
	IDL – E2	Throttle valve fully open		Infinity
		Throttle valve fully closed		0
	PSW – E2	Throttle valve fully open		0
		Throttle valve fully closed		Infinity
	THA – E2	Intake air temp. 20°C (68°F)		2,000 – 3,000
THW – E2	Coolant temp. 80°C (176°F)		200-400	
G1 _ G (-)	Cold		185-265	
Fuel cut	w/ Vehicle speed 0 km/h and coolant and coolant temp. 80°C (176°F) Fuel cut rpm Fuel return rpm		2,300 rpm 1,700 rpm	

Torque Specifications

Part tightened		N·m	kgf·cm	ft· lbf
Fuel line	Union bolt type	29	300	22
	Flare nut type	30	310	22
Fuel pump bracket x Fuel tank		2.9	30	26 in.-lbf
Fuel inlet pipe x Fuel tank		2.9	30	26 in.-lbf
Fuel evaporation vent tube x Fuel tank		1.5	15	13 in.-lbf
Fuel breather tube x Fuel tank		1.5	15	13 in.-lbf
Fuel tank band x Body		39	400	29
Fuel pressure regulator x Delivery pipe		9.3	95	82 in.-lbf
Delivery pipe x Cylinder head		15	150	11
Throttle body x Intake manifold		22	220	16

MFI AND SFI SYSTEMS (3S-GTE)

Specifications

Fuel pressure regulator	Fuel pressure at no vacuum	226 – 265 kPa (2.3 – 2.7 kgf/cm ² , 33 – 38 psi)	
Cold start injector	Resistance Fuel leakage	2–4Ω One drop or less per minute	
Injector	Resistance Injection volume Difference between each injector Fuel leakage	2–4Ω 95–120 cm ³ (5.8 – 7.3 cu in.) per 15 sec. 5 cm ³ (0.3 cu in.) or less One drop or less per minute	
Air flow meter	Resistance VS – E2 VC–E2 THA – E2 at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (104°F) at 60°C (140°F)	200 – 600 Ω (Measuring plate fully closed) 20 – 1,200 Ω (Measuring plate fully open) 200 – 400 Ω 10, – 20,kΩ 4–7 kΩ 2–3 Ω 0.9 – 1.3 kΩ 0.4 – 0.7 Ω	
Throttle position sensor	Clearance between stop screw and lever	Between terminals	Resistance
	0 mm 0 in. 0.50 mm 0.020 in. 0.70 mm 0.028 in. Throttle valve fully open —	VTA – E 2 IDL – E2 IDL – E2 VTA – E 2 VC–E2	0.47 – 6.1 kΩ 2.3 kΩ or less Infinity 3.1 – 12.1 kΩ 3.9 – 9.0 kΩ
Throttle opener	Setting speed	900 – 1,900 rpm	
ISC valve	Resistance +B – RSC or RSO	19.3 – 22.3 Ω	

Specifications (Cont'd)

Cold start injector time switch	Resistance STA – STJ below 10°C (50°F) above 25°C (77°F) STA – Ground	30 – 50Ω 70 – 90Ω 30 – 90Ω		
Solenoid resistor	Resistance +B – No.10, No.20, No.30 or No.40	4 – 6Ω		
Fuel pump resistor	Resistance	Approx. 0.73Ω		
T–VIS VSV	Resistance	33–39 Ω		
Turbocharging pressure VSV	Resistance	24 – 30 Ω		
EGR VSV	Resistance	33 – 39Ω		
Water temp. sensor	Resistance at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (104°F) at 60°C (140°F) at 80°C (176°F)	10–20 kΩ 4–7 kΩ 2 – 7 kΩ 0.9 – 1.3 kΩ 0.4 – 0.7 kΩ 0.2 – 0.4 kΩ		
EGR gas temp. sensor (CALIF. only)	Resistance at 500C (112°F) at 100°C (212°F) at 150°C (302°F)	69 – 89 kΩ 11–15kΩ 2–4 kΩ		
Oxygen sensor heater	Resistance	5.1 – 6.3 Ω		
ECU	HINT: • Perform all voltage and resistance measurements with the ECU connected. • Verity that the battery voltage is 11 V or above with the ignition switch ON.			
	Voltage			
	Terminals	Condition	STD voltage (V)	
	+ B _ E1 +B 1	IG SW ON	10–14	
	BATT – E1		10–14	
	IDL – E2	IG SW ON	Throttle valve open 4.5–5.5	
	VTA – E2		Throttle valve fully closed (Throttle opener must be cancelled first) 0.1 – 1.0	
			Throttle valve fully open 3.2–4.2	
				4.5–5.5
	VC – E2			Measuring plate fully closed 3.7–4.3
	VS – e2			Measuring plate fully open 0.2–0.5
			Idling 1.6 – 4.1	
			3,000 rpm 1.0–2.0	

Specifications (Cont'd)

ECU (cont'd)		Voltage (cont'd)		
		Terminals	Condition	
	No.1 No.2 _ E01 No.3 E02 No.4	IG SW ON		10–14
	THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1–3
	THW – E2		Coolant temp. 80°C (176°F)	0.1–1.1
	STA – E1	Cranking		6–14
	IGT – E1	Cranking or idling		0.8–1.2
	RSC _ RSO	IG SW ON	Engine ECU connectors disconnected	8–14
	W – E1	No trouble ("CHECK" engine warning light off) and engine running		10–14
	PIM – E2	IG SW ON		2.5–4.5
	AC – E1	IG SW ON	Air conditioning ON	8–14
	ACT – E1		Air conditioning ON	4–6
	TVIS – E1		Throttle valve fully closed	2.0 or less
			Throttle valve open	10–14
w/ Regular unleaded gasoline	TVIS – E1	Idling		2.0 or less
w/ Premium unleaded gasoline		4,200 rpm or more		10–14
	TE1 – E1	IG SW ON	Check connector TE1 – E1 not connected	10–14
			Check connector TE1 – E1 connected	0.5 or less
	Resistance			
	Terminals	Condition		STD resistance (Ω)
	IDL – E2	Throttle valve fully open		Infinity
		Throttle valve fully closed		2,300 or less
	VTA – E2	Throttle valve fully open		3,100 – 12,100
		Throttle valve fully closed		470–6,100
	VC – E2	—		390–9,000
	VS – e2	Measuring plate fully closed		200–600
		Measuring plate fully open		20–1,200
	THA – E2	Intake air temp. 20°C (68°F)		2,000 – 3,000
	THW – E2	Coolant temp. 80°C (176°F)		200–400

Specifications (Cont'd)

ECU (cont'd)	Resistance (cont'd)		
	Terminals	Condition	STD resistance (Ω)
	G1 _ G2 G (–)	Cold	125–190
	NE – G (–)	Cold	155–240
	RSC _ +B RSO +B1		19.3–22.3
Fuel cut	Fuel return rpm		1,600 rpm

Torque Specifications

Part tightened		N–m	kgf–cm	ft–lbf
Fuel line	Union bolt type	29	300	22
	Flare nut type	30	310	22
Fuel pump x Fuel tank		2.9	30	26 in.–lbf
Fuel sender gauge x Fuel tank		1.5	15	13 in.–lbf
Fuel evaporator bent tube x Fuel tank		1.5	15	13 in.–lbf
Fuel inlet pipe x Fuel tank		2.9	30	26 in.–lbf
Fuel tank band x Body		22	220	16
Cold start injector x Intake manifold		5.9	60	52 in.–lbf
Cold start injector pipe x Cold start injector		12	125	9
Cold start injector pipe x Delivery pipe		12	125	9
Fuel pressure regulator x Delivery pipe		29	300	22
Injector cover x Delivery pipe		7.8	80	69 in.–lbf
Fuel inlet hose x Delivery pipe	Bolt	7.8	80	69 in.–lbf
	Union bolt	29	300	22
Delivery pipe x Cylinder head		19	195	14
Fuel inlet hose x Fuel filter		29	300	22
Throttle body x Intake manifold		19	195	14
Intake air connector stay x Throttle body		19	195	14
Intake air connector stay x Cylinder head		7.8	80	69 in.–lbf
Intake air connector x Throttle body		19	195	14

MFI AND SFI SYSTEMS (5S-FE)

Specifications

Fuel pressure regulator	Fuel pressure at no vacuum	265 – 304 kPa (2.7 – 3.1 kgf/cm ² , 38 – 44 psi)	
Injector	Resistance Injection volume Difference between each injector Fuel leakage	Approx. 13.8 kΩ 49 – 59 cm ³ (3.0 – 3.6 cu i n.)/15 sec. 5 cm ³ (0.31 cu in.) or– less One drop or less per minute	
Throttle position sensor	Clearance between stop screw and lever	Between terminals	Resistance
	0 mm 0 in. 0.50 mm 0.020 in. 0.70 mm 0.028 in. Throttle valve fully open —	VTA – E2 IDL–E2 IDL – E2 VTA – E2 VC–E2	0.2 – 5.7 kΩ 2.3 kΩ or less Infinity 2.0 – 10.2 kΩ 2.5 – 5.9 kΩ
Throttle opener	Setting speed	1,300 – 1,500 rpm	
ISC valve	Resistance +B – ISCC or– ISCO	19.3 – 22.3Ω	
A/C idle–up VSV	Resistance	30 – 34Ω	
EGR VSV	Resistance	33–3992	
Water temp. sensor	Resistance at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (1040F) at 60°C (140°F) at 80°C (1761F)	10–20 kΩ 4–7 kΩ 2–3 kΩ 0.9 – 1.3 kΩ 0.4 – 0.7 kΩ 0.2 – 0.4 kΩ	
Intake air temp. sensor	Resistance at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (104°F) at 60°C (140°F) at 80°C (176°F)	10–20 kΩ 4–7 kΩ 2–3 kΩ 0.9 – 1.3 kΩ 0.4 – 0.7 kΩ 0.2 – 0.4 kΩ	
EGR gas temp. sensor (CALIF. only)	Resistance at 50°C (112°F) at 100°C (212°F) at 1500C (3020F)	69 – 89 kΩ 11–15kΩ 2–4 kΩ	
ECU	HINT: • Perform all voltage and resistance measurements with the ECU connected. • Verity that the battery voltage is 11 V or above with the ignition switch ON.		
	Voltage		
	Terminals	Condition	STD voltage (V)
	+B – E1 +B1	IG SW ON	10–14

Specifications (Cont'd)

ECU (cont'd)	Voltage (cont'd)			
	Terminals	Condition		STD voltage (V)
	BATT – E1			10–14
	IDL – E2	IG SW ON	Throttle valve open	8–14
	VTA – E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.8–1.2
			Throttle valve fully open	3.2–4.2
	PIM – E2	IG SW ON		3.3–3.9
	VC – E2			4.5–5.5
	No.10 _ E01 No. 20 E02			10–14
	THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1.9–2.9
	THW – E2		Coolant temp. 80°C (176°F)	0.1–1.1
	STA – E1	Cranking		6–14
	IGT – E1	Cranking or idling		0.8–1.2
	ISCC _ E1 ISCO	IG SW ON	Engine (& ECT) ECU connectors disconnected	8–14
	W – E1	No trouble ("CHECK" engine warning light off) and engine running		10–14
	A/C – E 1	IG SW ON	Air conditioning ON	8–14
	ACT – E1		Air conditioning ON	4.5–5.5
	ACA – E1		Air conditioning ON	2 or less
	TE1 – E1		Check connector TE1 – E1 not connected	10–14
	NSW – E1		Check connector TE1 – E1 connected	1 or less
			Neutral start switch P or N range	0 – 2
			Ex. neutral start switch P or– N range	6–14
	B/K – E1	Stop light SW ON (Brake pedal depressed)		10–14
	Resistance			
	Terminals	Condition		STD resistance (Ω)
	IDL – E2	Throttle valve fully open		Infinity
		Throttle valve fully closed (Throttle opener must be cancelled first)		2,300 or– less
	VTA – E2	Throttle valve fully open		2,000 – 10,200
		Throttle valve fully closed (Throttle opener must be cancelled first)		200–5,700

Specifications (Cont'd)

ECU (cont'd)	Resistance (cont'd)		
	Terminals	Condition	STD resistance (Ω)
	VC – E2		2,500 – 5,900
	THA – E2	Intake air temp. 20°C (68°F)	2,000 – 3,000
	THW – E2	Coolant temp. 80°C (176°F)	200–400
	G + – G –	Cold	185–265
	NE + – NE –	Cold	370–530
	ISCC _ +B ISCO +B I		19.3–22.3
Fuel cut	Fuel return rpm		1,500 rpm

Torque Specifications

Part tightened		N·m	kgf·cm	ft·lbf
Fuel line	Union bolt type	29	300	22
	Flare nut type	30	310	22
Fuel pump bracket x Fuel tank		2.9	30	26 in.-lbf
Fuel inlet pipe x Fuel tank		2.9	30	26 in.-lbf
Fuel evaporation bent tube x Fuel tank		1.5	15	13 in.-lbf
Fuel sender gauge x Fuel tank		1.5	15	13 in.-lbf
Fuel breather gauge x Fuel tank		1.5	15	13 in.-lbf
Fuel tank band x Body		39	400	29
Cold start injector x Intake manifold		9.3	95	82 in.-lbf
Cold start injector pipe x Cold start injector		18	180	13
Cold start injector pipe x Delivery pipe		19	195	14
Fuel pressure regulator x Delivery pipe		5.4	55	48 in.-lbf
Fuel return pipe x Fuel pressure regulator		18	180	13
Delivery pipe x Cylinder head		13	130	9
Fuel pulsation damper x Delivery pipe		34	350	25
Throttle body x Intake manifold		19	195	14

COOLING SYSTEM

Specifications

Engine coolant capacity		See page A-2
Radiator cap	Relief valve opening pressure STD	74 – 103 kPa (0.75 – 1.05 kgf/cm ² , 10.7 – 14.9 psi)
	Limit	59 kPa 0.6 kgf/cm ² 8.5 psi
Thermostat	Valve opening temperature	80 – 84°C 176 –183°F
	Valve lift at 95°C (203°F)	8 mm (0.31 in.) or more

Torque Specifications (4A-FE)

Part tightened	N·m	kgf·cm	ft·lbf
Engine block x Drain plug	34	350	25
Water pump x Cylinder block	15	150	11
Water inlet pipe x Water pump	20	200	14
Water inlet pipe x Cylinder block	13	130	9
Water inlet x Water inlet housing	9.3	95	82 in.-lbf

torque Specifications (3S-GTE and 5S-FE)

Part tightened		N·m	kgf·cm	ft·lbf
Engine block x Drain plug		25	250	18
Water pump x Water pump cover	3S-GTE	9.3	95	82 in.-lbf
	5S-FE	8.8	90	78 in.-lbf
Water pump x Cylinder block	3S-GTE	7.8	80	69 in.-lbf
	5S-FE	9.3	95	82 in.-lbf
Water by-pass pipe x Water pump cover	3S-GTE	12	120	9
	5S-FE	9.3	95	82 in.-lbf
Idler pulley bracket x Cylinder block (3S-GTE)		19	195	14
Idler pulley bracket x Alternator bracket (3S-GTE)		19	195	14
Water inlet x Water pump	3S-GTE	7.8	80	69 in.-lbf
	5S-FE	9.3	95	82 in.-lbf

LUBRICATION SYSTEM

Specifications

Engine oil capacity			See page A-29	
Oil pressure	at idle at 3,000 rpm		29 kPa (0.3 kgf/cm ² , 4.3 psi) or more 245 – 490 kPa (2.5 – 5.0 kgf/cm ² , 36 – 71 psi)	
Oil pump (4A-FE)	Body clearance	STD	0.080 – 0.180 m m	0.0031 – 0.0071 in.
		Limit	0.20 m m	0.0079 i n.
	Tip clearance	STD	0.025 – 0.085 m m	0.0010 – 0.0033 in.
		Limit	0.35 m m	0.0138 in.
	Side clearance	STD	0.025 – 0.085 m m	0.0010 – 0.0033 in.
		Limit	0.10 mm	0.0039 in.
Oil pump (3S-GTE and 5S-FE)	Body clearance	STD	0.100–0.160 mm	0.0039 – 0.0063 in.
	Tip clearance	Limit	0.20 m m	0.0079 in.
		STD	0.040 – 0.160 m m	0.0016 – 0.0063 in.
		Limit	0.20 m m	0.0079 i n.

Torque Specifications (4A-FE)

Part tightened	N– m	kgf–cm	ft–lbf
Engine pan x Drain plug	34	350	25
Oil pump x Cylinder block	21	218	16
Oil strainer x Cylinder block	9.3	95	82 in.–lbf
Oil strainer x Oil pump	9.3	95	82 in.–lbf
Oil pan x Cylinder block	4.9	55	43 in.–lbf
Oil pan x Oil pump	4.9	55	43 in.–lbf
Oil pan x Rear oil seal retainer	4.9	50	43 in.–lbf
Oil dipstick guide x Water pump	9.3	95	82 in.–lbf

Torque Specifications (3S-GTE and 5S-FE)

Part tightened		N·m	kgf·cm	ft·lbf
Engine pan x Drain plug		39	400	29
Oil pump body cover x Oil pump body		8.8	90	78 in.-lbf
Oil pump x Cylinder block	3S-GTE	7.8	80	69 in.-lbf
	5S-FE	9.3	95	82 in.-lbf
Oil strainer x Cylinder block		5.4	55	48 in.-lbf
Oil strainer x Oil pump		5.4	55	48 in.-lbf
Oil pan x Cylinder block		5.4	55	48 in.-lbf
Oil pan x Oil pump		5.4	55	48 in.-lbf
Stiffener plate x Cylinder block		37	380	27
Stiffener plate x Transaxle case		37	380	27
Oil cooler bracket x Cylinder block (3S-GTE)		7.8	80	69 in.-lbf
Oil cooler x Oil cooler bracket (3S-GTE)		78	800	58
Water by-pass pipe x Oil cooler (3S-GTE)		12	120	9
Water by-pass pipe x Oil cooler bracket (3S-GTE)		18	180	13
Oil cooler x Cylinder block (5S-FE)	Relief valve	78	800	58
	Nut	7.8	80	69 in.-lbf
Oil nozzle x Cylinder block		9.1	93	81 in.-lbf

IGNITION SYSTEM

Ignition timing	10° BTDC @ idle (w/ Terminals TE1 and E1 connected)		
Firing order	1-3-4-2		
Spark plug	See page A-2		
High-tension cord	Resistance	25 kΩ per cord	
Ignition coil	Primary coil resistance (Cold) 4A-FE 3S-GTE and 5S-FE Secondary coil resistance (Cold)	1.1-1.7 Ω 0.3-0.6 Ω 9-15Ω	
Distributor	Air gap Signal generator (pickup coil) resistance (Cold) 4A-FE 3S-GTE G1 - G (-) G2 - G (-) NE -G (-) 5S-FE G (+)- G (-) NE (+) - NE (-)	0.2 - 0.4 m m	0.008-0.016 in. 185 - 265 Ω 125 - 190 Ω 125 - 190 Ω 155 - 240 Ω 185 - 265 Ω 370 - 530 Ω

STARTING SYSTEM

Starter	Rated voltage and output power		12 V 1.0 kW		12V1.4kW,12V1.6 kW
	No-load characteristic	Current	90 A or less at 11.5 V		←
		rpm	3,000 rpm or more		3.500 rpm or more
	Brush length	STD	13.5 mm	0.531 in.	15.5 mm 0.610 in.
		Limit	8.5 mm	0.335 in.	10.0 mm 0.394 in.
	Commutator				
	Outer diameter	STD	30.0 mm	1.181 in.	←
		Limit	29.0 mm	1.142 in.	←
	Undercut depth	STD	0.6 mm	0.024 in.	←
		Limit	0.2 mm	0.008 in.	←
	Circle runout	Limit	0.05 mm	0.0020 in.	←
	Spring installed load		18–24 N		←
			(1.79 – 2.41 kgf, 3.9–5.3 lbf)		←

CHARGING SYSTEM

Drive belt tension			See page A-2	
Battery specific gravity when fully charged at 20°C (68°F)			1.25–1.27	
Alternator	Rated output		12 V–70 A, 12V–80 A	
	Rotor coil resistance		2.8–3.092	
	Slip ring diameter	STD	14.2 – 14.4 mm	0.559 – 0.567 in.
		Limit	12.8 mm	0.504 in.
	Brush exposed length	STD	10.5 mm	0.413 in.
Alternator regulator		Limit	1.5 mm	0.059 in.
	Regulating voltage	at 25°C (77°F)	13.9 – 15.1 V	
		at 115°C (239°F)	13.5 – 14.3 V	

LUBRICANT

Item		Capacity			Classification
		Liters	US qts	Imp. qts	
Manual transaxle oil	C52	2.6	2.7	2.3	API GL-4 or GL-5 SAE 75W-90
	E150F	5.2	5.5	4.6	API GL-5 Viscosity: Above -18°C (0°F) SAE90 Below -18°C (0°F) SAE80W
Automatic transaxle fluid A241 E Dry fill Drain and refill A243L Dry fill Drain and refill		8.0	8.5	7.0	ATF DEXRON® II
		3.3	3.5	2.9	
		7.7 3.3	8.1 3.5	6.8 2.9	
Differential oil		1.1	1.2	1.0	API GL-5 Hypoid gear oil Viscosity: Above -18°C (0°F) SAE 90 Below -18°C (0°F) SAE80W-90
Power steering fluid	Total	0.8	0.85	0.70	ATF DEXRON® II
Brake fluid		—			SAE J1703 or FMVSS No. 116 DOT3

LUBRICANT

Item			Capacity			Classification
			Liters	U S qts	Imp. qts	
Engine oil	4A-FE	Dry fill	3.7	3.9	3.3	API grade SG, multigrade fuel-efficient and recommended viscosity oil
		Drain and refill				
		w/ Oil filter change	3.2	3.3	2.8	
		w/o Oil filter change	3.0	3.1	2.6	
	3S-GTE	Dry fill	4.3	4.5	3.8	
		Drain and refill				
		w/ Oil filter change	3.9	4.1	3.4	
		w/o Oil filter change	3.6	3.8	3.2	
	5S-FE (w/ Oil cooler)	Dry fill	4.6	4.9	4.0	
		Drain and refill				
		w/ Oil filter change	4.2	4.4	3.7	
		w/o Oil filter change	3.8	4.0	3.3	
	5S-FE (w/o Oil cooler)	Dry fill	4.5	4.8	4.0	
		Drain and refill				
		w/ Oil filter change	4.1	4.3	3.5	
		w/o Oil filter change	3.7	3.9	3.3	

STEERING

Specifications

Steering	Steering wheel freeplay	30 mm (1.18 in.) or less
Steering column	Pawl stopper (Tilt steering column) Mark 1 or A 2 or B 3 or C 4 or D 5 or E 6 or F 7 or G (Auto tilt away steering column) Mark 1 or A 2 or B 3 or C 4 or D 5 or E 6 or F 7 or G Collar (Auto tilt away steering column)	12.68 – 12.74 mm 0.4992 – 0.5016 in. 12.61 – 12.67 mm 0.4965 – 0.4988 in. 12.54 – 12.60 mm 0.4937 – 0.4961 in. 12.47–12.53 mm 0.4909 – 0.4933 in. 12.40 – 12.46 m m 0.4882 – 0.4906 in. 12.33 – 12.39 mm 0.4854 – 0.4878 in. 12.26 – 12.32 mm 0.4827 – 0.4850 in. 12.68 – 12.74 mm 0.4992 – 0.5016 in. 12.61 – 12.67 mm 0.4965 – 0.4988 in. 12.54 – 12.60 mm 0.4937 – 0.4961 in. 12.47 – 12.53 mm 0.4909 – 0.4933 in. 12.40 – 12.46 mm 0.4882 – 0.4906 in. 12.33 – 12.39 mm 0.4854 – 0.4878 in. 12.26 – 12.32 mm 0.4827 – 0.4850 in. 12.75 – 12.85 mm 0.5020 – 0.5059 in. 12.55 – 12.65 mm 0.4941 – 0.4980 in. 12.35 – 12.45 mm 0.4862 – 0.4902 in. 12.15 – 12.25 mm 0.4783 – 0.4823 in. 11.95 – 12.05 mm 0.4705 – 0.4744 in. 11.75 – 11.85 mm 0.4610 – 0.4665 in. 11.55 – 11.65 mm 0.4547 – 0.4587 in. 11.25 – 11.35 mm 0.4429 – 0.4468 in.
Power steering (On-vehicle inspection)	Drive belt tension New belt Used belt Maximum rise of oil level Oil pressure at Idle speed (4A-FE engine) (5S-FE engine) (3S-GTE engine)	441 ± 667 N 45 – 68 kgf 100 – 150 l bf 265 ± 441 N 27 – 45 kgf 60 – 100 l bf Below 5 mm 10.20 in.) 6,865 kPa (70 kgf/cm ² , 996 psi) 7,355 kPa (75 kgf/cm ² , 1,068 psi) 7,845 kPa (80 kgf/cm ² , 1,138 psi)
	Steering effort USA CANADA	39 N 4 kgf 8.8 l bf 6.9 N –m 70 kgf–cm 61 in.–lbf

Specifications (Cont'd)

(Power steering pump)	Rotor shaft bushing oil clearance STD		0.01 – 0.03 mm	0.0004 – 0.0012 in.
	Maximum		0.07 mm.	0.0028 in.
	Vane plate to rotor groove clearance		0.03 mm	0.0012 in.
	Vane plate Minimum height		8.0 mm	0.315 in.
	Minimum thickness		1.77 mm	0.0697 in.
	Minimum length		14.97 mm	0.5894 in.
	Vane plate length Rotor and cam ring mark			
		None		
		1		
		2		
Flow control valve spring length		3	14.996 – 14.998 mm	0.59039 – 0.59047 in.
		4	14.994 – 14.996 mm	0.59032 – 0.59039 in.
	STD		14.992 – 14.994 mm	0.59024 – 0.59032 in.
	Minimum		14.990 – 14.992 mm	0.59016 – 0.59024 in.
			14.988 – 14.990 mm	0.59008 – 0.59016 in.
			38 mm	1.49 in.
			36 mm	1.42 in.
	Pump rotating torque		0.3 N-m (2.8 kgf-cm, 2.4 in.-lbf) or less	
(Gear housing)	Steering rack runout	Maximum	0.3 mm	0.012 in.
	Bearing preload (turning)		0.4 – 0.6 N-m (4.5 – 6.5 kgf-cm, 3.9 – 5.6 in.-lbf)	
	Total preload (turning)		0.9 – 1.2 N-m (9 – 12 kgf-cm, 7.8 – 10.4 in.-lbf)	

Torque Specifications

Steering column (Non-tilt steering column)	Part tightened		N-m	kgf-cm	ft-lbf
	Main shaft x Intermediate shaft		35	360	26
(Tilt steering column)	Tilt lever assembly installation bolt		2.9	30	26 in.-lbf
	Tilt pawl set nut		5.9	60	52 in.-lbf
	Tilt lever retainer set nut		15	150	11
	Compression spring bushing bolt		7.8	80	69 in.-lbf
	Main shaft x Intermediate shaft		35	360	26
(Auto tilt away steering column)	Tilt memory bolt		6.4	65	56 in.-lbf
	Tilt lever bracket installation bolt		2.9	30	26 in.-lbf
	Tilt lever retainer set nut	14 mm	15	150	11
		8 mm	2.9	30	26 in.-lbf
	No. 2 tilt memory plate set nut	14 mm	15	60	11
		10 mm	5.9	80	52 in.-lbf
	Compression spring bushing bolt		7.8	360	69 in.-lbf
	Main shaft x Intermediate shaft		35		26
(Removal and installation)	Main shaft x Steering wheel		35	360	26
	Steering wheel pad (w/ airbag)		7.4	75	65 in.-lbf
	Column tube x Body		25	260	19
	Column tube attachment x Brake pedal bracket		12	120	9
	Universal joint		35	360	26

Torque Specifications (Cont'd)

Power steering (Power steering pump)	Part tightened	N-m	kgf-cm	ft. lbf
	Pressure port union x Front housing	69	700	51
	Air control valve x Pump housing	36	370	27
	Drive pulley nut	43	440	32
	Reservoir tank x Housing (4A-FE)	13	130	9
	12 mm	41	420	30
	14 mm	69	700	51
			130	9
	Union bolt (4A-FE)	13	525	38
	Suction port union x Pump housing (5S-FE)	51	400	29
	Pressure tube joint bolt	39	440	32
	PS pump x Bracket (4A-FE)	43	400	29
	PS pump x Bracket (5S-FE, 3S-GTE) Upper	39	370	27
	Lower	36	550	36
	Pressure tube union nut (5S-FE, 3S-GTE))	49	1,550	152
	Tie rod end x Knuckle arm	152	740	29
	Lower crossmember R H and LH sides	39		
	Center			
(Gear housing)	Control valve housing x Rack housing	31	315	23
	Bearing guide lock nut	56	570	41
	Rack guide spring cap lock nut	56	570	41
	Rack x Rack end	72	730	53
	Turn pressure tube x Gear housing	20	200	14
	Tie rod end lock nut	56	570	41
	Gear housing x Body	59	600	43
	Gear housing x Return line	44	450	33
	Gear housing x Pressure line	44	450	33
	Center member x Body	52	530	38
	Tie rod end x Knuckle arm	49	500	36
	Control valve shaft x Universal joint	35	360	26
	Lower crossmember	152	1,550	112
	Tube clamp set bolt	13	130	9
	Engine x Engine mounting	77	790	57
	Engine mounting x Center member			
	4A-FE			
	5S-FE, 3S-GTE			
	Lower crossmember R H and LH sides	64	650	47
	Center	73	740	54
	Engine mounting x Insulator	152	1,550	112
	Exhaust front pipe x Manifold	39	400	29
	Exhaust front pipe x Center pipe	87	890	64
	Propeller shaft x Intermediate shaft	62	630	46
	Stabilizer bar x Link	43	440	32
	Stabilizer bar bracket	74	750	54
		35	360	26
		18	180	13